

21SLA2

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Problem

Which positive integers n make the equation

$$\sum_{i=1}^n \sum_{j=1}^n \left\lfloor \frac{ij}{n+1} \right\rfloor = \frac{n^2(n-1)}{4}$$

true?

Solution. We claim that this only happens when $n+1$ is a prime.

Observe that

$$\sum_{i=1}^n \sum_{j=1}^n \left\lfloor \frac{ij}{n+1} \right\rfloor = \frac{n^2(n-1)}{4} \iff \sum_{i=1}^n \sum_{j=1}^n \left\{ \frac{ij}{n+1} \right\} = \frac{n^2}{2}$$

Consider the sum

$$\sum_{i=1}^n \sum_{j=1}^n \left(\left\{ \frac{ij}{n+1} \right\} + \left\{ \frac{(n+1-i)j}{n+1} \right\} \right)$$

If $n+1 \nmid ij$ for all $i, j \leq n$ then

$$\sum_{i=1}^n \sum_{j=1}^n \left(\left\{ \frac{ij}{n+1} \right\} + \left\{ \frac{(n+1-i)j}{n+1} \right\} \right) = n^2$$

If $\exists i, j \neq n$ such that $n+1 \mid ij$ then

$$\sum_{i=1}^n \sum_{j=1}^n \left(\left\{ \frac{ij}{n+1} \right\} + \left\{ \frac{(n+1-i)j}{n+1} \right\} \right) < \sum_{i=1}^n \sum_{j=1}^n 1 = n^2$$

So the only case left is that of $n+1 \nmid ij$ for all $i, j \leq n$. It is easy to show that this only happens when $n+1$ is prime.